

MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)
ROTATING MACHINERY INSPECTION REPORT - VDU UNIT 2

=====

Equipment ID: MRPL-COMP-201-A

Equipment Description: VDU Wet Gas Multi-Stage Centrifugal Compressor

Operating Unit: VDU-2 (Vacuum Distillation Overhead Section)

Inspection Date: 2026-09-08

Inspector: J. B. Hegde (Senior Reliability Specialist, API 617 Certified)

OBSERVATIONS & VIBRATION SPECTRUM ANALYSIS:

- Thrust Bearing Active Face Temperature: 94 deg C (Max Limit: 88 deg C).
- Stage-1 Radial Shaft Vibration (Probe X): 6.8 mm/s RMS (Alarm Limit: 4.5 mm/s).
- Stage-1 Radial Shaft Vibration (Probe Y): 6.4 mm/s RMS.
- Sub-synchronous Vibration (0.43X Running Speed): Oil whirl / hydrodynamic instability detected.
- Lube Oil Differential Pressure (Delta-P): Dropped to 0.72 bar (Normal range: 1.2 - 1.6 bar).
- Dry Gas Seal Nitrogen Buffer Flow: Elevated at 18.5 Nm³/h (Normal baseline: < 10 Nm³/h).

ROOT CAUSE HYPOTHESIS:

Lube oil viscosity degradation combined with severe labyrinth seal contamination leading to hydrodynamic journal bearing destabilization and incipient axial thrust collar scrubbing.

MAINTENANCE DIRECTIVE & SAFETY CLASS: CRITICAL (LEVEL 1)

1. Immediate load reduction on VDU-2 by 25% to throttle suction volumetric flow.
2. Emergency shutdown within 24 hours to prevent catastrophic rotor-stator contact.
3. Replace tilting-pad thrust bearing shoes and perform high-pressure lube filter swap.
4. Overhaul Dry Gas Seal (DGS) cartridge #1 and restore N₂ buffer differential.